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ABSTRACT

This paper presents preliminary results of a study aimed at specification, selection and preliminary design of subsystems required to transform maneuvering, reentry body technology into a piloted, high performance Spaceplane, potentially capable of operating throughout cislunar space. This study, under contract to the U.S. Air Force, is being conducted by SRI International, Hamilton Standard and several other aerospace contractors. Potential Spaceplane applications include placement, support, servicing and retrieval of satellites, and a variety of other military missions. The emphasis in this paper is on key issues relating to integration of the crewman into the vehicle. Issues discussed include environmental control/life support, cabin and spacesuit pressure levels and methods of heat rejection. Human factors issues relating to displays and controls, and prevention of the bends are also presented. Finally, major Spaceplane integration issues, such as center-of-gravity control, and payload and recovery system location are also addressed.

KEYWORDS

Spaceplane, manned spacecraft, crew accommodation, Environmental Control & Life Support, spacecraft design, Extravehicular Activity.

SPACEPLANE REQUIREMENTS

Military missions in space in the future will require a vehicle with a high degree of autonomy, flexibility, maneuverability, responsiveness, survivability and cost effectiveness. Such a vehicle must possess the following characteristics:

- Independent of extensive ground control, monitoring, tracking and communications.
- Capable of space rescue operations.
- High degree of maneuverability in space.
- External carry of large payloads.
- Flexible mission profiles, launch schedules, reentry conditions, and recovery.

- Possibility of multiple launch sites and survivable launch.
- Capable of supporting an astronaut in extravehicular activity.

Non-military manned spacecraft including the Space Shuttle have not possessed these requirements, having been designed for a very limited set of missions and mission profiles. Further, today's manned space vehicle programs and concepts have precipitated the commonly-held perception that the economics, technology and safety of man in space will force the continuation of non-military characteristics into the future. There is interest, therefore, to provide the military with an effective high performance space vehicle such that the resulting system satisfies the required military mission characteristics and provides the capability to support the space resources of the United States at all orbital altitudes. While this country currently relies heavily on unmanned satellites as vital elements in command, control, communications, intelligence, surveillance, reconnaissance and warning, manned vehicles can provide distinct advantages. A manned spacecraft can be less vulnerable to attack by anti-satellite weapon systems, can perform multiple missions on an as-required basis by being adaptive and able to react to changing mission conditions and can operate autonomously. Therefore, advancement of manned space vehicle research and technology must be done to complement ongoing work in military satellite, sensor and communication technologies.

Analysis sponsored by the Department of Defense has shown that limitations exemplified above for manned space vehicles and unmanned satellites are not inherent in a new type of manned military space vehicle termed the High-Performance Spaceplane (HPSP). An initial configuration of the HPSP is depicted in Figure 1. The straked-wing, conical, reentry vehicle is approximately twenty-five feet long and has a base diameter of fifty inches. For cislunar and other high velocity-change missions, the external propulsion module shown with cryogenic propellant tanks is attached around and extends over the aft section of the reentry body. An internal, storable propellant, propulsion system is

standard in the basic vehicle to provide propulsion for many types of missions in low-to-medium altitude orbits. External tankage can be attached for increased velocity over and above the internal propulsion system capability. The HPSP is compatible with Space Shuttle, aircraft launch and a family of expendable launch vehicles. Land recovery of HPSP by a controllable parachute is baseline.

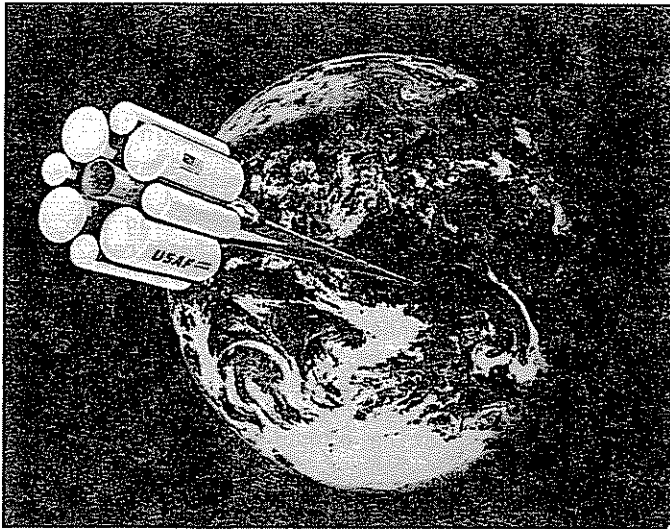


Fig. 1 - HPSP concept

The HPSP provides the potential for performing and adding missions throughout cislunar space and the upper atmosphere comprehensively and to do so with truly military mission characteristics. HPSP could be used to defend satellites, gather reconnaissance information and conduct communications and intelligence missions. The vehicle could also fill an important role by servicing geosynchronous-orbit communications satellites, since extravehicular activity (EVA) by the pilot is included in the baseline concept.

This paper discusses issues relating to integrating a crewman into an HPSP. Considering required subsystems, the environmental control and life support system (EC/LSS) are the subsystems that are expected to have the greatest impact on the HPSP design. Results of studies to define basic EC/LSS requirements consistent with mission and HPSP vehicle definition are presented. Specifically, the EC/LSS components discussed include:

- Environmental Control Subsystems
- EVA Equipment
- Crew Provisions

Food and Water Supply
Couches
Waste Management
Consumables
Flammability Control

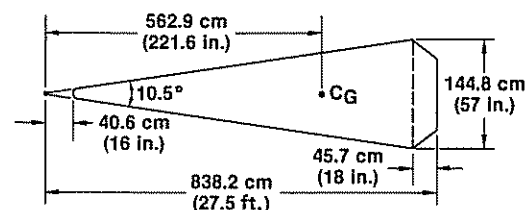
- Controls and Displays
- Support Equipment

In each case, subsystem and vehicle interface considerations are oriented toward achieving a totally integrated Environmental Control and Life Support Subsystem for HPSP. It should be noted that many of the mission-related issues which had an effect on the trade studies discussed herein cannot be presented due to their sensitivity.

INTEGRATION ISSUES

The vehicle currently under consideration uses a reentry body airframe based on the Sandia Winged Energetic Reentry Vehicle, Experimental (SWERVE), under development by Sandia National Laboratories, as a maneuvering reentry body. This conical vehicle, characterized in Figure 2, presents a significant challenge in terms of integrating the numerous systems and subsystems into the allowable volume while maintaining the specified CG limits required for aerodynamic stability. An HPSP will require the following systems to become an operable vehicle:

- Thermal Protective System (TPS)
- Lift Control Surfaces
- Environmental Control/Life Support System (EC/LSS)
- Substructure
- Ballast
- Electric Power Subsystem
- Life Support
- Avionics and Communications
- Controls and Displays
- Recovery System
- Propulsion/Attitude Control



Overall length — 792.5 cm (26 ft.)
Included cone angle — 10.5 deg
Wall thickness — 7.6 cm (3 in.)
Desired CG — 562.9 cm (221.6 in.) (from cone apex)

Fig. 2 - Operational spaceplane characteristics

- Pilot/Couch
- Internal Payload/External Payload

Looking at the vehicle defined in the above-referenced figure, it is clear that a judicious use of volume inboard of the structure must be made to accommodate packaging of all subsystems. In addition, the subsystems must be capable of 0-g and 1-g operation, as well as being able to withstand acceleration loads during launch, reentry and synergistic plane changes (for orbital maneuvers). The challenge in subsystem equipment integration is to achieve a high packaging density to keep the vehicle small. Therefore, selection of equipment must involve technology which will result in low volume and weight. In addition to the issues related to packaging, control of the center of gravity in both radial and axial directions is important to vehicle dynamic performance. Therefore, placement of equipment to minimize CG shifts as consumables are utilized is required. A final, but most important, issue to be considered is positioning of the pilot to provide adequate mobility to accomplish required tasks and withstand all acceleration and deceleration loads without physiological impairment. These issues affect the method of integrating equipment as well as crewman accommodations and will be addressed further in the last section of this paper.

VEHICLE LAYOUT

The basic geometry of the high performance Spaceplane concepted is shown in Figure 3. The conical vehicle is approximately 7.92m (26 ft) long with an included cone angle of 10.5 degree and a dry weight in the range of 1814-2268 kg (4000-5000 lb). Using the on-board propulsion system and control surfaces, the vehicle would have capability for orbit changes, together with atmospheric maneuvering. The nose cone of the vehicle is hinged to fold back on-orbit to provide payload access or viewing.

Integration of various subsystem equipment is indicated in Figure 3, which shows in greater detail systems/hardware which interface or directly support the crew.

Major constraints in configuring the vehicle layout were maximum vehicle weight and length, specified center-of-gravity, folding nose, vehicle conical geometry, minimum specified propulsion system ΔV and maximum G-load profile. In addition, an attempt was made to maintain mission flexibility and pilot/vehicle safety while maintaining a reasonable probability of mission success. The folding nose feature influenced hardware packaging to the extent that equipment involving fluid coolant or propellant was not packaged in the nose section to avoid the need for flexible fluid lines. To minimize ballast required to trim for the desired CG, the conical shape of the vehicle dictated that where possible, heavy components be packaged forward, with light components toward the rear of the vehicle. Major consumables such as water and propellant were balanced about the CG in such a way that their consumption during the mission would result in minimal shift in overall vehicle CG. Volume for ballast was located at the vehicle nose tip where it would have the longest lever arm to the vehicle CG. The volume allocated for payload was located near the hinge point so that, on-orbit with the nose folded back, the payload would be accessible and have viewing capability. The main recovery system was located near the vehicle CG to minimize shroud line runs to the vehicle load bearing suspension ring. The pilot was located toward the rear of the vehicle to provide adequate volume for task mobility and in an upright position to offer protection against the physiological effects due to axial directed G-loads during HPSP launch and flight maneuvers. Major HPSP integration issues considered are summarized in Figure 4.

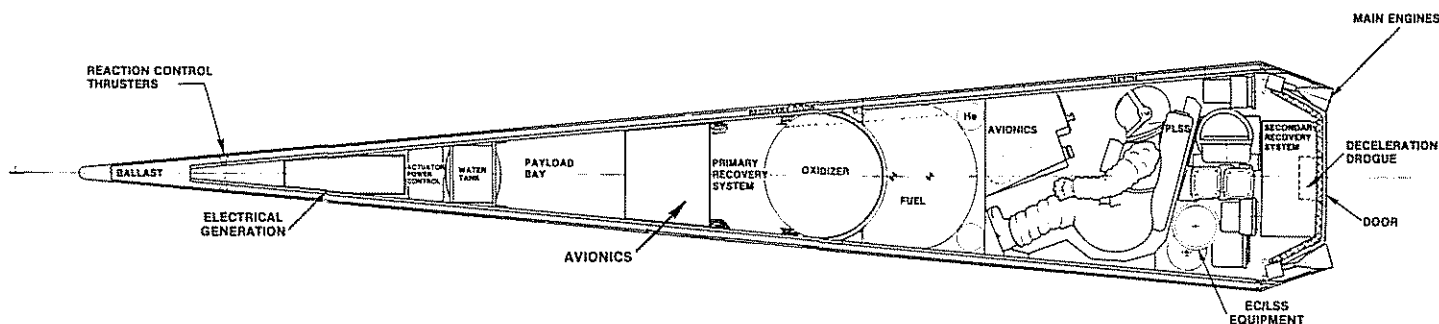


Fig. 3 - Vehicle layout

- Desired center-of-gravity
- Vehicle geometry/characteristics
- Subsystem geometry/characteristics
- Maintainability requirements
- Packing density
- Pilot positioning
- Pilot/vehicle interfaces
- EVA egress
- Consumables location
- Component location constraints
- G-load profiles
- Payload requirements
- Recovery system deployment

Fig. 4 - Integration issues

In follow-on HPSP studies, effort will be made to further reduce total vehicle weight. The benefit is that for a given total vehicle weight, the reduction in hardware weight can be allocated for an increase in propellant which increases the total vehicle ΔV available. In addition, emphasis will be on increasing subsystem and overall integration packing density to further reduce vehicle length. Packing density P_D is defined as

$$P_D = \frac{\text{Total subsystems volume}}{\text{Total vehicle packaging volume}} \times 100$$

For example, in the Apollo program, the life support backpack in the spacesuit system achieved a $P_D = 45\%$. In the Space Shuttle program, the new Portable Life Support System (PLSS) in the EMU has a $P_D = 80\%$. Results of initial HPSP integration studies indicate a preliminary $P_D = 62\%$, excluding volume allocated for payloads. Key considerations in future work to increase HPSP packing density are summarized in Figure 5.

- Optimize subsystem functional schematic
- Minimize component volume
- Conformal component packaging
- Combine components
- Optimize fluid line sizes
- Custom designs

Fig. 5 - Packaging density considerations

CREW ACCOMMODATIONS

The general guideline adopted in this study was to define an HPSP achievable in the near term with lowest program development cost. Details of the specific approaches recommended will be discussed in subsequent sections.

The various interfaces between the vehicle and the HPSP crewman are shown in their most general form in Figure 6. A useful way to view these parameters is in terms of inputs to and outputs from the crewman in the two different modes of operation, i.e., Intra-vehicular Activity (IVA) and Extravehicular Activity (EVA). The first mode, IVA, involves the seated crewman interacting with HPSP cockpit systems, while the second has the crewman on EVA interacting with the Extravehicular Mobility Unit (EMU) and other systems (e.g., satellites) external to HPSP.

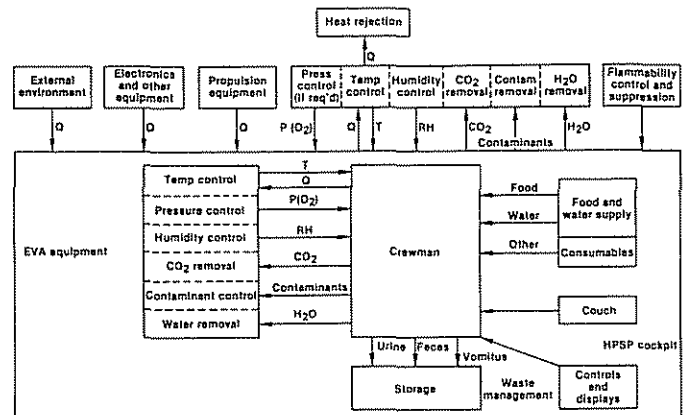


Fig. 6 - HPSP crew interfaces

In either mode of operation, as indicated in Figure 6, the crewman requires basic consumable inputs of oxygen for breathing and water to meet metabolic requirements. He generates metabolic by-product outputs in the form of moisture, carbon dioxide, heat and various other liquid, gas and solid waste products which HPSP systems must accommodate. The moisture, carbon dioxide, gas contaminants (e.g., odors) and heat must be continuously removed from the pilot's atmosphere. Oxygen to replace that consumed must also be provided. Metabolic heat plus that generated from HPSP electrical and propulsion systems and solar wall heating of HPSP that would raise the pilot's environmental temperature need to be controlled via a heat rejection device. Solid or liquid waste products generated by the crewman must be accommodated by collector and storage systems. The second mode of operation to consider is where the pilot is on EVA, supported by the EMU spacesuit and life support subsystems. All of the pilot IVA requirements discussed above must now be met by the EMU Portable Life Support System (PLSS) independent from HPSP life support systems. With the exception of having to improve present spacesuit waste management capabilities, the requirements are basically the same as those presently provided by the Space Shuttle Extravehicular Mobility Unit (EMU).

A couch/restraint system which interfaces with the EMU is required for pilot comfort and for protection during HPSP launch and flight maneuvers. HPSP controls and displays, designed with appropriate human factors considerations taken into account, are required to provide the pilot with subsystem and mission performance status and the capability to effect control of systems based on the analysis of performance data.

ENVIRONMENTAL CONTROL/LIFE SUPPORT SYSTEM (EC/LSS)

The design reference mission (DRM) assumed for this study was based on a one man crew with a mission capability of up to 24-hour duration. Schematically shown in Figure 7 is the equipment required to support the HPSP crewman in a space environment. In the cabin, the pilot suited in the EMU would utilize the vehicle EC/LSS, using short umbilicals with the EMU serving as emergency IVA back-up. The EMU liquid cooling garment (LCG) is connected to the vehicle cooling loop for removal of the metabolic heat load. The coolant selected for use in the vehicle loop and, therefore, the LCG, is water to eliminate potential hazards to the pilot from toxic coolants, such as the Freons. Water is then circulated through a series of cold

plates for cooling attached components, such as avionics, payloads, vehicle control surface actuators and any other equipment requiring cooling. Upon leaving these cold plates, the coolant flows through a heat rejection device before being returned to the LCG. The choice of heat sink is discussed below.

Breathing air and ventilation cooling are provided to the crewman in a loop which circulates gas from the man through a lithium hydroxide canister for carbon dioxide removal. The air is then cooled in the heat sink and passes through a rotary water separator for removal of condensed moisture. It is then re-oxygenated with make up oxygen, prior to being returned to the man from the high pressure oxygen supply. The stored high pressure oxygen is also used to pressurize the water and ammonia, which are used for heat rejection to provide overall temperature control. As discussed below, an unpressurized cabin (i.e., the crewman completely suited at all times) was selected for HPSP. The three key trade studies performed in the EC/LSS area involved the selection of an unpressurized crew compartment, selection of water for the circulating coolant and evaluation of concepts for the heat rejection device.

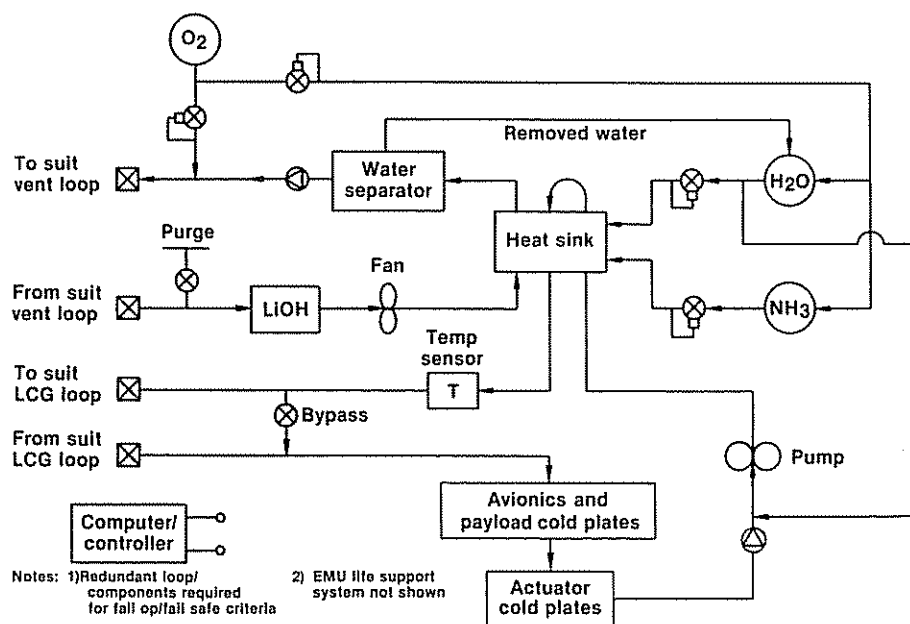


Fig. 7 - Spaceplane EC/LSS schematic

Pressurized Vs Unpressurized Cockpit

As shown in Figure 8, Hamilton Standard traded-off an unpressurized cockpit versus a cockpit pressurized with either oxygen, an oxygen/nitrogen mixture or an inert gas. Key issues included impact on volume and weight of such parameters as vehicle wall thickness, seals and hatch mechanisms, stored gas for leakage make-up or cabin repressurization and pilot volume requirements. Also considered was the possibility of fire in the cockpit, and the potential impact on the mission caused by leakage or crewman incapacitation. A key issue in this trade was whether the crewman could function for the entire 24-hour full mission while fully suited. It was concluded that, with improved waste management (i.e., solid and liquid wastes and vomitus collection), a crewman could perform the one-day mission fully suited with helmet and gloves. While it would also be desirable to allow the crewman the additional freedom provided by doffing gloves and helmet, an unpressurized cabin was selected as the baseline configuration for the HPSP for the following reasons:

- reduced vehicle size/weight
- reduced fabrication cost
- less stringent sealing requirements
- reduced maintenance cost
- minimization of fire hazard if require a high O₂ content
- simplified operations
- little or no mission impact from a vehicle cabin leak.

Key considerations

- Vehicle size/weight
- Cockpit leakage
- Flammability control
- Mission profile impact
- Cabin repressurization

Options

- Unpressurized cockpit
- Pressurized cockpit
 - O₂ atmosphere
 - O₂/N₂ atmosphere
 - Inert atmosphere

Fig. 8 – Pressurized vs. unpressurized trade study

Major considerations in considering the various options on which these conclusions are based are summarized in Figure 9.

Advantages

- Reduced weight (~ 9.1 kg [20 lbs] for thinner walls)
- Reduced volume (no helmet/glove donning/ storage)
- Reduced leakage
- No fire control in vacuum
- No mission impact due to significant leak

Disadvantages

- Pilot suited at all times
- Requires improved solid waste management
- Requires vehicle repressurization

Fig. 9 – Unpressurized cockpit considerations

Heat Transport

Two options considered for heat management throughout the Spaceplane vehicle to cool equipment and remove metabolic and environmental loads were primarily circulating coolant or heat pipes. Trade-off criteria include cooling capability, volume and weight impact and performance limitations. Heat pipes were rejected primarily due to their performance degradation when subjected to varying g-loads, because of the fixed requirement for circulation of coolant through the spacesuit LCG and because of difficulties in interfacing heat pipes with the cold plates which provide equipment cooling. Water was selected as the coolant to be used in the circulating coolant heat transport loop because of its compatibility with existing suit components, its excellent thermal characteristics and its lack of toxicity. However, the choice of water will necessitate protecting coolant loop components from freezing.

Heat Rejection

Figure 10 summarizes the trade study performed to select the heat rejection device to be utilized by HPSP. An estimated average heat rejection rate of 4000 Btu/hr for HPSP is composed of the following sources:

<u>Source</u>	<u>Heat Load-Watts (Btu/Hr)</u>
Metabolic	293.1 (1000)
Avionics and Communications	146.6 (500)
Electrical Power Equipment	293.1 (1000)
Payload	146.6 (500)
Other	293.1 (1000)

- Key considerations**
- Size and weight
 - Performance limitations
 - Mission impact
 - Control requirements
- Options**
- Radiator
 - Thermal capacitor (phase change material)
 - Evaporation: Flash evaporator
Spray boiler
Stacked evaporator
- Assumptions**
- Required heat rejection of 4000 BTU/hr for 24 hour mission
 - Steam exhaust acceptable for detectability

Fig. 10. - Thermal control trade study

Use of a radiator was eliminated from further consideration for the short duration HPSP mission because of the volume and weight impact, 0.21m³ (7.5 ft³) with 8.3m² (89 ft²) surface area and over 45.4 kg (100 lbs), its inability to operate in the atmosphere during a synergistic plane change and its impact on mission profiles by forcing certain orientations. Also of importance is the complexity added by a deployment/retraction mechanism and the increased detectability posed by the radiator's highly reflective surface. As mission length increases, the advantage of a radiator in reducing expendables could become extremely significant (i.e., 1.8 kg/hr (4 lbs/hr) of water evaporated for heat rejection). Similarly, a phase change device, using heat of fusion instead of heat of vaporization, was rejected due to the large weight of phase change material required to reject 1172 watts (96,000 Btu/day). Therefore, evaporative cooling was selected as the means for HPSP heat rejection.

Due to its high heat of vaporization and density, water was selected for use above 30.5 km (100,000 feet) and on-orbit, while ammonia is required during descent below about 30.5 km (100,000 feet) as a result of its lower saturation pressures. Hamilton Standard evaluated devices similar to the Space Shuttle Orbiter Flash Evaporator Subsystem (FES) and Water Spray Boiler (WSB) and also a derivative device concept called a "stacked evaporator", Figure 11. The stacked evaporator was selected for HPSP application for reasons indicated in Figure 12. The FES was rejected since a cylindrical shape is required to conform to the required spray nozzle pattern. This configuration requires excessive volume as compared to other approaches. There would also be a need to changeover from H₂O to NH₃ in the same core during descent (as the HPSP goes below 100,000 ft) and there would be need for exhaust pressure control during descent to prevent NH₃ evaporation temperature from becoming too low and freezing the coolant loop. The WSB was rejected for these latter two reasons plus the fact that it has a higher degree of carryover in a zero-g environment, which would require additional expendables.

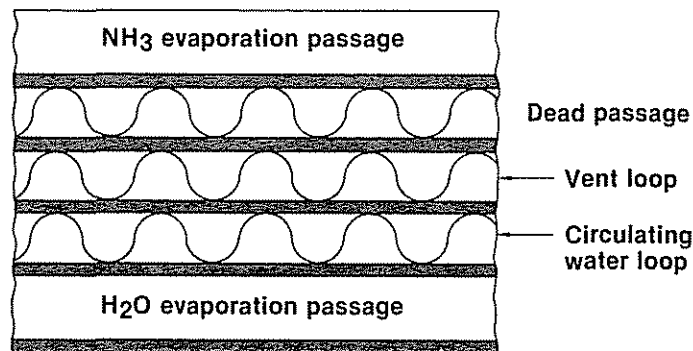


Fig. 11 - Stacked evaporator concept

Advantages

- All heat rejection provided by single device
- Dead passage eliminates exhaust pressure control and NH₃ leakage
- Lightweight device (approx. 4.54 kg (10 lbs))
- Little wasted volume (approx. 1475 cm³ (90 in³) vs. 3932 cm³ (240 in³) for flash evaporator)

Disadvantages

- Development required

Fig. 12 - Stacked evaporation considerations

EXTRAVEHICULAR ACTIVITY

As discussed previously the EC/LSS selected for HPSP includes an unpressurized vehicle cabin with the pilot in a spacesuit throughout all mission phases. The spacesuit and life support system selected for HPSP is a 55.2 kPa (8 psi) derivative of the Shuttle EMU with self-contained portable life support system (PLSS). For IVA, the crewman would be supported by vehicle life support systems and consumables via an umbilical. To conduct EVA, the pilot would disconnect the EC/LSS umbilicals at the suit, which would activate the PLSS. Since previous experience has shown that management of EVA umbilicals required more than one crewman, the use of the EMU for EVA was selected. The advantages of HPSP capability to conduct EVA are many. For example, EVA supported by simple hand tools could be used to conduct satellite service, assist in payload deployment and retrieval and be used for space rescue operations.

The EC/LSS candidate spacesuit options considered viable for HPSP, together with principal discriminators considered in the selection process, are indicated in Figure 13. It was apparent from considering candidate HPSP mission profiles that the existing Shuttle EMU, while satisfying most EC/LSS and EVA requirements would require modification. A modified 27.6 kPa (4 psi) EMU was then defined which incorporated minimum, required

modifications to the existing Shuttle EMU to meet HPSP mission requirements. Finally, a 55.2 kPa (8 psi) EMU was defined which incorporates several additional features beyond those of the modified 27.6 kPa (4 psi) EMU to provide desired HPSP mission quick response and flexibility. A comparison of the three spacesuit system configurations that were studied is made in Figure 14. The results of the comparison study were that the existing Shuttle EMU would not satisfy HPSP requirements without the modifications defined in Figure 14 for the modified 27.6 kPa (4 psi) EMU. Therefore, these two candidates were very close in configuration. It was concluded further that the HPSP spacesuit system should be the 55.2 kPa (8 psi) EMU system.

Options

- Shuttle EMU
- Modified 27.6 kPa (4 psi) EMU
- 55.2 kPa (8 psi) EMU

Discriminators

- Weight/volume
- Mobility
- Prebreathe requirement
- Crew safety
- Mission impact
- Mission scenario

Fig. 13 – Spacesuit considerations

Feature	Shuttle EMU	Modified 27.6 kPa (4 psi) EMU	55.2 kPa (8 psi) EMU
Total pressure	28.3 kPa (4.1 psi)	28.3 kPa (4.1 psi)	55.2 kPa (8 psi)
Prebreathe	3 hours O ₂	3 hours O ₂	None
Gloves and joints	Existing	Existing	New gloves and joints
Umbilical provision	SCU — not adequate	SCU — modify vent and water loops in PLSS	Yes
EVA capability	7 hours nominal	7 hours nominal	7 hours nominal
UCD capacity	950cc	2000cc	2000cc
Solid waste management	None	Solid collectors New helmet required	Yes
Water and food	595g (21 oz) water 227g (8 oz) food stick	1021g (36 oz) water 454g (16 oz) food stick	1021g (36 oz) water 454g (16 oz) food stick
1-G mobility	Limited	Limited	Good with detachable PLSS
Dry weight	109 kg (240 lb)	109 kg (240 lb)	113 kg (250 lb)
Consumables	0.55 kg (1.22 lb) oxygen 4.54 kg (10 lb) water 4.21 kg (9.3 lb) battery 1.13 kg (2.5 lb) LiOH	Same	Same
Couch interface	PLSS accommodated by couch	PLSS integral part of upper couch	PLSS integral part of upper couch and detachable
Structure	Not designed for 3G's with man	PLSS structural modifications	Designed for 3G's with man
Comfort	Not designed for 3G's with man	HUT padding and neck ring cushioning	Modified upper torso

Fig. 14 – Candidate EMU comparisons

The reason for a 55.2 kPa (8 psi) EMU is that this choice of spacesuit operating pressure eliminates the requirement for oxygen pre-breathe prior to going EVA which, with an unpressurized cabin, is effectively what the pilot would do upon launch regardless of the launch mode. Even if the cabin were to be pressurized, with a 27.6 kPa (4 psi) EMU serving as backup, prebreathe would be required since a cabin leak early in the mission would subject the pilot to the bends. The choice of 55.2 kPa (8 psi) pressure is based on results of recent prebreathe studies which indicate that a crewmember acclimated to a 101.4 kPa (14.7 psi) ambient or the

Space Shuttle 101.4 kPa (14.7 psi) environment could immediately go EVA at a suit pressure (pure oxygen) of 55.2 kPa (8 psi) without experiencing denitrogenization bend problems. Presently, going EVA in the 27.6 kPa (4 psi) EMU spacesuit system requires approximately a 3-hour, pure oxygen pre-breathe. The elimination of prebreathe is consistent with the quick response capability required in military missions. The most significant development impact of elevated spacesuit pressure is the need for development of new gloves and suit joints to maintain crew mobility and hand dexterity for performing both IVA and EVA tasks.

Preliminary results of one recent prebreathe study are shown in Figure 15 and which are of interest in the context of an 55.2 kPa (8 psi) spacesuit for HPSP. Figure 15 assumes first that a crewman, acclimated to an earth-like environment at some time prior to zero, enters a 55.2 kPa (8 psi) spacesuit of pure oxygen, and, after 10 minutes, purges the suit of the nitrogen removed from the lung cavities during breathing. At time zero the crewmember then continues tissue denitrogenization in the spacesuit environment of essentially pure oxygen at a total pressure of 55.2 kPa (8 psi). Analysis results, shown in Figure 15, indicate the elapsed time required in the 55.2 kPa (8 psi) spacesuit to safely depressurize in a single step to lower suit pressures of 41.4, 34.5 and 27.6 kPa (6, 5 and 4 psi). The advantage of a step-wise spacesuit depressurization provision are that based on certain mission scenarios, the crewmember could safely reduce suit pressure prior to going EVA, which would improve work mobility, and reduce spacesuit joint wear.

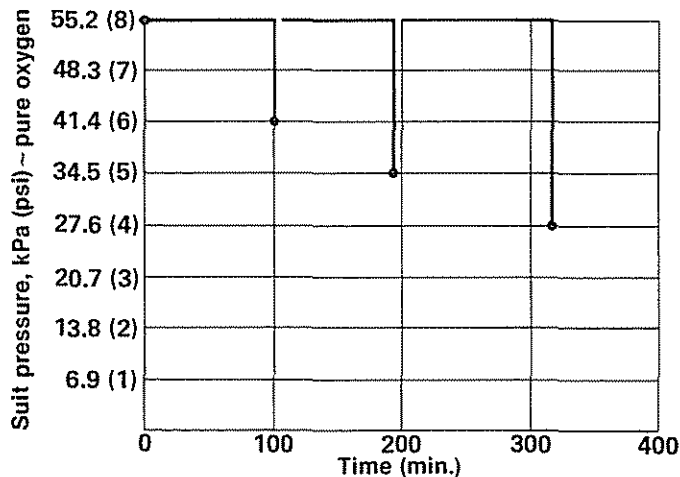


Fig. 15 - Variable suit pressure as a function of time

Since this type of scenario is desirable for HPSP operation, NASA and the Air Force have expressed an interest in pursuing such technology.

A diet adequate to sustain the pilot throughout the 24-hour HPSP mission should provide approximately 3000 to 4000 Calories of energy, with adequate quantities of proteins, fats, carbohydrates, minerals and vitamins. Exact caloric requirements for pilots vary based on the actual body weight, age, height, and level of activity for the specific mission. Adequate water, contained in a drink bag, and food sticks would be integrated into the EMU suit and accessible from within the EMU.

Waste management consists of provisions for urine, vomitus, and feces collection and storage. The urine collection device would have increased capacity from that of the existing Shuttle EMU to provide for the 24-hour mission duration. Preliminary conceptual design work has been done on devising insuit solid waste collectors to provide for safe hygienic containment and storage of feces (beyond the current baseline diaper arrangement) and vomitus. Concepts being studied make use of filter collection devices and suction sources.

As conceived, the Portable Life Support System (PLSS) will essentially form the upper couch structure which will minimize couch volume and weight. The design of the flight couch proposed must reflect the vehicle inside envelope, the EMU features, EVA equipment, vehicle controls and displays and other human factors.

Padding for crew comfort will be provided within the upper spacesuit torso itself. The buttocks and legs will be supported by the lower couch which will permit movement and repositioning for comfort and vehicle control. Provisions for adjustment include tilting the couch to aid in launch positioning, on-orbit visual observations and to facilitate EVA activities.

The interaction of the spacesuited pilot and the vehicle while engaged in IVA and EVA requires consideration with regard to controls and displays which must be designed to be accessible, operable and communicative. Any HPSP manual controls (e.g., levers, buttons, switches) must be designed to be compatible with the pilot's pressurized glove geometry and dexterity limits. The use of voice control and optical system heads-up-displays integrated into the pilot's helmet system is being studied and offers many advantages both in performance, flexibility and weight and volume savings. Another EVA consideration is pilot-induced vehicle dynamics. While the pilot is restrained in his couch on IVA, analysis indicates that changes in momentum resulting from torso or head motions can be effectively nulled by the on-board attitude control system momentum wheels. However, when free from the couch restraint system, for example, during EVA egress through the hatch, pilot motions can induce vehicle moments on the order of 4.07 joule (36 in-lb). For a vehicle weighing 2268 kg (5000 lbs) and geometries studied, this could result in vehicle yaw, pitch and roll rates in excess of 30 degrees/minute. Ideally, should the pilot impart dynamics to the vehicle when on EVA, vehicle dynamics would be nulled when the pilot again stabilizes through physical contact with the vehicle.

However, a net angular displacement of the vehicle from its original attitude would have occurred in the interim. The approach taken may be to allow vehicle attitude perturbations during EVA. When the pilot reenters the vehicle following EVA, the vehicle guidance system would be used to reestablish vehicle attitude. In addition, when on EVA, the pilot would take along the cockpit control panel in the form of a computer-driven virtual image.

CONCLUSIONS

The integration of man into a maneuvering reentry vehicle presents a challenge that can be met with a combination of existing Shuttle EMU hardware and components which are derivative of other life support equipment. Good design practice, coupled with in-depth knowledge of manned systems, will allow this integration into such a quick response space-

craft to be accomplished. This capability will be required in the future as we increase our utilization of space.

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